
Modern Algebra 1

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Problem 1

Solution. (a) We know $\mathbb{Z}_n \times \mathbb{Z}_m \cong \mathbb{Z}_{nm}$ if and only if $\gcd(n, m) = 1$. Since $\gcd(3, 9) = 3 \neq 1$, the groups are not isomorphic.

(b) By the theorem mentioned in part (a), we know $\mathbb{Z}_6 \times \mathbb{Z}_5 \cong \mathbb{Z}_{30}$, and $\mathbb{Z}_3 \times \mathbb{Z}_{10} \cong \mathbb{Z}_{30}$, since $\gcd(5, 6) = 1$ and $\gcd(3, 10) = 1$. Since $\cong$ is an equivalence relation on the class of groups, by the transitive property we have $\mathbb{Z}_6 \times \mathbb{Z}_5 \cong \mathbb{Z}_3 \times \mathbb{Z}_{10}$.

(c) Notice that D_4 is nonabelian. Consider some $(a_1, b_1), (a_2, b_2) \in \mathbb{Z}_2 \times \mathbb{Z}_4$. Since $\mathbb{Z}_2$ and $\mathbb{Z}_4$ are abelian, we have $a_1 + a_2 = a_2 + a_1$ and $b_1 + b_2 = b_2 + b_1$. It thus follows that

$$(a_1, b_1)(a_2, b_2) = (a_1 + a_2, b_1 + b_2) = (a_2 + a_1, b_2 + b_1) = (a_2, b_2)(a_1, b_1).$$

Therefore, $\mathbb{Z}_2 \times \mathbb{Z}_4$ is abelian. Since D_4 is nonabelian, there exists no isomorphism.

(d) For an isomorphism to exist, there must exist a bijection, and thus $|D_6 \times \mathbb{Z}_2| = |A_4|$. We know $|D_6 \times \mathbb{Z}_2| = |D_6| |\mathbb{Z}_2| = 12 \cdot 2 = 24$. Additionally, we know $|A_4| = 4!/2 = 24/2 = 12$. Since $24 \neq 12$, there exists no isomorphism. ■

Problem 2

Solution. Let $G = \langle g \rangle$, and let $H = \langle h \rangle$. Suppose $G \times H$ is cyclic. Notice that the order of $G \times H = |G| |H|$. It follows that there exists some $(a, b) \in G \times H$ such that $|(a, b)| = |G| |H|$. Since $|(a, b)| = \text{lcm}(|a|, |b|)$, we have

$$|H| |G| = \frac{|a| |b|}{\gcd(|a|, |b|)}.$$

The maximum value of $|a|$ for some $a \in G$ is $|G|$, since G is cyclic. Similarly, the maximum value of $|b| \in H$ is $|H|$. Therefore, the maximum value of $|a| |b|$ is $|H| |G|$. Note that if $|a| |b| < |H| |G|$, then

$$\frac{|a| |b|}{\gcd(|a|, |b|)} < |H| |G|,$$

since $\gcd(|a|, |b|) \geq 1$ for all a and b , and thus dividing by it cannot increase the number. Therefore, we fix $|a| |b| = |H| |G|$, since this is the only value of the item's order that will generate the group. It then follows that

$$\frac{|H| |G|}{\gcd(|H|, |G|)} = |H| |G| \implies \gcd(|H|, |G|) = 1.$$

We now show the converse. Suppose $|G|$ and $|H|$ are relatively prime. We know $|g| = |G|$ and $|h| = |H|$ since g and h generate their respective groups, and thus we have

$$\frac{|h||g|}{\gcd(|h|, |g|)} = \frac{|H||G|}{\gcd(|H|, |G|)} = \frac{|H||G|}{1} = |H||G| = |H \times G|.$$

■

Problem 3

Solution. Suppose HK is a group. Since HK is a group, for all $h_1k_1 \in HK$, there exists some $h_2k_2 \in HK$ such that $(h_1k_1)^{-1} = h_2k_2$. It follows that

$$\begin{aligned} (h_1k_1)^{-1} &= h_2k_2 \\ \implies \left((h_1k_1)^{-1}\right)^{-1} &= (h_2k_2)^{-1} \\ \implies h_1k_1 &= k_2^{-1}h_2^{-1}. \end{aligned}$$

Notice that $k_2^{-1}h_2^{-1} \in KH$. Therefore, for all $hk \in HK$, we have $hk \in KH$, and thus $HK \subseteq KH$. Now let $kh \in KH$. Since K and H are groups, k has an inverse in K and h has an inverse in H . Therefore, $(kh)^{-1} = h^{-1}k^{-1}$. Since $(kh)^{-1} \in G$, and thus it has an inverse, it follows that $kh = (h^{-1}k^{-1})^{-1}$. Since $h^{-1}k^{-1} \in HK$ and HK is a group, it follows that $(h^{-1}k^{-1})^{-1} \in HK$, and thus $kh \in HK$. Therefore, $KH \subseteq HK$. Thus, $HK = KH$.

Now we show the converse. Let $h_1k_1, h_2k_2 \in HK$. Since $HK = KH$, it follows that there exists some $k_3h_3 \in KH$ such that $h_2k_2 = k_3h_3$. We then have $h_1k_1h_2k_2 = h_1k_1k_3h_3$. Since K is a group, $k_1k_3 \in K$. Notice that $k_1k_3h_3 \in KH$. Since $HK = KH$, there exists some $h_4k_4 \in HK$ such that $h_4k_4 = k_1k_3h_3$. It follows that $h_1k_1k_3h_3 = h_1h_4k_4$. Notice that since H is a group, $h_1h_3 \in H$, and thus $h_1h_4k_4 \in HK$. Therefore, HK is closed under the group operation. Since G is a group, and thus $HK \subseteq G$, associativity holds for all hk . Since H and K are both groups, the identity $e \in H, K$. It follows that $ee = e \in HK$, and thus there exists an identity. Finally, let $hk \in HK$. Since $hk \in G$, we know $(hk)^{-1} = k^{-1}h^{-1}$. Since $k^{-1}h^{-1} \in KH$ and $KH = HK$, there exists some $h'k' \in HK$ such that $k^{-1}h^{-1} = h'k'$. Therefore, $(hk)^{-1} = h'k' \in HK$. Therefore, HK is a group. ■

Problem 4

Solution. (a) Since H and K are groups, $H \times K$ is a group. Since $H \subseteq \mathbb{Z}$ and $K \subseteq \mathbb{Z}$, then $H \times K \subseteq \mathbb{Z} \times \mathbb{Z}$. Therefore, $H \times K$ is a subgroup of $\mathbb{Z} \times \mathbb{Z}$.

(b) Let $H, K \leq \mathbb{Z}$ be subgroups, and suppose $H \times K$ is cyclic. It follows that $H \times K$ is generated by some element (n, m) . Since $\mathbb{Z}$ is cyclic, by the fundamental theorem of cyclic groups we know H, K are cyclic. It follows that $H = \langle a \rangle$ and $K = \langle b \rangle$. Thus, $H \times K = a\mathbb{Z} \times b\mathbb{Z}$. Notice that since (n, m) generates $H \times K$, and since obviously $(a, b) \in H \times K$, there exists some k such that $k(n, m) = (a, b)$. It follows that $kn = a$ and $km = b$. Notice again that $2b \in b\mathbb{Z}$, and thus $(a, 2b) \in H \times K$. It follows that there exists some k' such that $k'(n, m) = (a, 2b)$. Since $k'n = a$, we have $k'n = kn$. Since $k'm = 2b$, we have $k'm = 2km$. If $n \neq 0$, we have $k' = k$, and thus $km = 2km$. It follows that $m = 0$ in the general case, since there are cases where $k \neq 0$. It follows that $H \times K = a\mathbb{Z} \times \{0\}$. If we suppose instead that $m \neq 0$, by the same logic we obtain $H \times K = \{0\} \times b\mathbb{Z}$. Assuming $n, m \neq 0$ fairly obviously gives the contradiction $1 = 2$, and assuming

$n, m = 0$ is simply a special case of the cases $n = 0$ or $m = 0$. Therefore, the cyclic subgroups $H \times K$ of $\mathbb{Z} \times \mathbb{Z}$ are those of the form $a\mathbb{Z} \times \{0\}$ or $\{0\} \times b\mathbb{Z}$.

(c) The set $\langle(1, 1)\rangle$ does not equal $H \times K$ for any $H, K \leq \mathbb{Z}$. Suppose for purpose of contradiction that $\langle(1, 1)\rangle = H \times K$ for some $H, K \leq \mathbb{Z}$. It follows that $1 \in H, K$. Since K is a group and thus closed under addition, $\langle 1 \rangle \subseteq K$, and since $\langle 1 \rangle = \mathbb{Z}$, we have $K = \mathbb{Z}$. By the same logic, we have $H = \mathbb{Z}$. Thus, $H \times K = \mathbb{Z} \times \mathbb{Z}$. Notice that $(2, 1) \in \mathbb{Z} \times \mathbb{Z}$, but $(2, 1) \notin \langle(1, 1)\rangle$. By contradiction, there exist no such $H \times K$ such that $\langle(1, 1)\rangle = H \times K$. ■

Problem 5

Solution. (a) Let some $(a_1, \dots, a_n) \in \mathbb{Z}_p^n \setminus \{e\}$, where e is the identity. Since p is prime, for all $a_i \in \mathbb{Z}_p \setminus \{0\}$, we know $|a_i| = p$. Additionally, if $a_i = 0$, we know $|a_i| = 1$, and $\text{lcm}(1, p) = p$. It follows that

$$|(a_1, \dots, a_n)| = \text{lcm}(|a_1|, \dots, |a_n|) = p,$$

since all a_i have either $|a_i| = p$ or $|a_i| = 1$. Therefore, for all $x \in \mathbb{Z}_p^n \setminus \{e\}$, we know $|x| = p$. Since $|\mathbb{Z}_p^n| = p^n$ and $|\{e\}| = 1$, there exist $p^n - 1$ elements of order p in $\mathbb{Z}_p^n$.

(b) Since only the identity has order 1, and all other elements of $\mathbb{Z}_p^n$ have order p , and for any prime, $p \geq 2$, we know there must be at least one nonzero in any subgroup of order p . It follows from this that each subgroup H of order p has at least one element h of order p , and thus H must be cyclic. Therefore, we are simply looking for the number of cyclic subgroups of $\mathbb{Z}_p^n$. Note that since there are $p^n - 1$ elements of order p in $\mathbb{Z}_p^n$, then there are $p^n - 1$ cyclic subgroups since there are $p^n - 1$ possible generators. However, these subgroups are not necessarily distinct. Since every nonzero element of a cyclic subgroup has order p , and since each cyclic subgroup is closed, then each cyclic subgroup has $p - 1$ generators. We divide the number of cyclic subgroups by the number of generators per subgroup to find that the number of subgroups of $\mathbb{Z}_p^n$ of order p is equal to

$$\frac{p^n - 1}{p - 1}.$$

(c) If $\langle x \rangle \langle y \rangle$ is an internal direct product of nontrivial cyclic subgroups of $\mathbb{Z}_p^n$, then we know by a theorem about internal direct products that $\langle x \rangle \langle y \rangle \cong \mathbb{Z}_p \mathbb{Z}_p \cong \mathbb{Z}_p^2$, since $\langle x \rangle, \langle y \rangle \cong \mathbb{Z}_p$, and the only subgroups of order p^2 are isomorphic to $\mathbb{Z}_p^2$ or $\mathbb{Z}_{p^2}$, and being isomorphic to $\mathbb{Z}_{p^2}$ implies there exists a generator of order p^2 , which we have shown does not exist in $\mathbb{Z}_p^n$. Additionally, $|\mathbb{Z}_p^2| = p^2$. Since $\mathbb{Z}_p^n$ is abelian, $\langle x \rangle \langle y \rangle$ will be an internal direct product if $\langle y \rangle \cap \langle x \rangle = \{e\}$. By our logic in part (b), we know there are $p^n - 1$ choices for x . We also know that $y \notin \langle x \rangle$, and $y \neq 0$, implying that there are $p^n - 1 - (p - 1) = p^n - p$ elements that we can choose from for y . Thus, there are $(p^n - 1)(p^n - p)$ possible choices for x, y . However, there are multiple generators for $\langle x \rangle$ and $\langle y \rangle$, so we must account for this.

Notice that any nonzero choice of x will generate $\langle x \rangle$, and any nonzero choice of y will generate $\langle y \rangle$. Thus, any nonzero choice of (x, y) will generate $\langle x \rangle \langle y \rangle$. Since there are $p^2 - 1$ nonzero elements of $\langle x \rangle \langle y \rangle$, there are $p^2 - 1$ choices of elements for x that will generate the group. We must then choose y in the group such that $y \notin \langle x \rangle$. Since the order of the group is p^2 and the order of $\langle x \rangle$ is p , there are $p^2 - p$ choices in y . By dividing out the amount by which we are overcounting, we find the number of subgroups of order p^2 is

$$\frac{(p^n - 1)(p^n - p)}{(p^2 - 1)(p^2 - p)}.$$